

Core Signatures and Inversions

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Abstract

The signature transform of a path produces a feature representation with several useful invariance and approximation properties, but the truncated signature is redundant. The standard remedy is the log signature, which has the dimension of the free nilpotent Lie algebra but is more expensive to compute. We introduce *core signatures*: subsets of the truncated signature with the same dimension as the log signature, identified algorithmically through the shuffle product and identified structurally with the Lyndon coordinates of the universal step- n free nilpotent Lie group $G^n(\mathbb{R}^d)$. We then use Chen's identity to invert a core signature into a piecewise linear path of two, three, or more segments that exactly reproduces the target signature up to a chosen truncation level, and frame the multi-segment inversion as a weighted nonlinear least-squares regression in signature space. We state a precise norm-equivalence conjecture that would identify this regression with a Sobolev projection on path space, supported by structural arguments and numerical evidence. Mathematica code and the latest companion notebook are available at <https://github.com/MarcosCarreira/CoreSignatures>.

Keywords: signatures; log signatures; Lyndon words; Witt formula; signature inversion; rough paths

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1 Introduction

Signatures are the standard feature representation for paths in rough path theory and in machine-learning applications to time series ([Lyons, Caruana and Lévy (2007), Chevyrev and Kormilitzin (2016), Fermanian (2021a)]). For a path $X : [a, b] \rightarrow \mathbb{R}^d$ of bounded variation, the signature $S(X)_{a,b}$ is the collection of all iterated integrals along X ; truncated at level m , it provides a finite-dimensional faithful encoding of X modulo tree-like equivalence. The truncated signature contains redundancies arising from the shuffle relations, and the standard tool for stripping them is the log signature, whose dimension matches that of the free nilpotent Lie algebra $\mathfrak{g}^m(\mathbb{R}^d)$ of step m on d generators ([Reizenstein (2015), Reizenstein (2016), Reizenstein (2018), Reizenstein and Graham (2020)]).

This paper develops two practical results. First, we identify a subset of the truncated signature — the *core signature* — with the same dimension as the log signature but consisting of actual signature entries. Once the identification with the Lyndon coordinates of the free nilpotent Lie group is in place, the core entries are read directly from $S(X)_{a,b}$ without any change of basis at computation time; the identification itself rests on Reutenauer’s basis theorem for the free Lie algebra, the Poincaré–Birkhoff–Witt isomorphism, and Radford’s theorem on the shuffle algebra (all classical). The core elements turn out to be exactly the entries indexed by Lyndon words on the alphabet $\{1, \dots, d\}$, and their count is given by Witt’s formula. Second, we use Chen’s identity to invert a core signature into a concatenation of two or three line segments in the time-ordered case, and into longer concatenations through a weighted least-squares regression in signature space whose solutions empirically coincide with the optimal piecewise-linear L^1 approximation of the target.

The closest related work is [Amendola, Friz and Sturmfels (2019)], who study the algebraic geometry of the universal variety $G^n(\mathbb{R}^d)$ and identify its dimension with the log-signature size, and [Pfeffer, Seigal and Sturmfels (2018)], who study polynomial and piecewise-linear paths up to level three. The coordinate-theoretic identification of the Lyndon-indexed signature entries with a transcendence basis of $\mathcal{O}(G^n(\mathbb{R}^d))$ is classical and follows from Reutenauer and Radford; our contribution in Section 3 is therefore the computational packaging — naming the subset, exhibiting an explicit shuffle-product elimination that produces it, tabulating the substitution rules Q_m , and using the resulting coordinates for Chen-based inversion — rather than a new coordinate theorem. The Gröbner-basis derivation of Section 3 is the algorithmic realisation of the abstract identification with Lyndon coordinates. For inversion algorithms in the smooth-path regime see [Fermanian et al. (2023)].

After our 2023 preprint, two computational-algebra developments deserve mention. [Amendola et al. (2025)] introduce a Macaulay2 package, `PathSignatures`, that computes signature varieties of piecewise polynomial paths and explicitly implements the Lyndon-shuffle representation underlying our substitution rules Q_m : their method `lyndonShuffle` expresses an arbitrary word in $T(\mathbb{R}^d)$ as a shuffle polynomial in Lyndon words, which is the same operation our Gröbner-basis derivation makes constructive. In Section 6 of that paper they use the term *core signature tensor* with a meaning different from ours: their core tensor is the signature of a base path D such that the paths of interest are obtained from D by linear transformations $A \circ D$, parametrising

families of signatures through the linear group action. Our core signature is a subset of the entries of a single signature, addressing redundancy *within* one path's signature. The two constructions are complementary rather than competing — theirs is suited to the algebraic geometry of signature varieties; ours to Chen-based inversion of individual signatures — but the terminology overlap is worth flagging. A parallel Julia/OSCAR implementation of related computational tools is described in [Riffo and Schmitz (2026)].

2 Signatures

2.1 Definitions

2.1.1 Paths and iterated integrals

A *path* is a continuous map $X : [a, b] \rightarrow \mathbb{R}^d$. Going forward we assume bounded variation and piecewise differentiability; the standard reference for the more general theory is [Lyons, Caruana and Lévy (2007)]. The interval $[a, b]$ may always be reparametrised to $[0, 1]$.

The signature of X is the formal series

$$S(X)_{a,b} = \left(1, S(X)_{a,b}^{i_1}, S(X)_{a,b}^{i_1, i_2}, \dots\right), \quad i_k \in \{1, \dots, d\}, \quad (1)$$

whose level- k entries are the iterated integrals

$$S(X)_{a,t}^{i_1, \dots, i_k} = \int_a^t \int_a^{t_k} \dots \int_a^{t_2} dX_{t_1}^{i_1} \dots dX_{t_k}^{i_k}. \quad (2)$$

We work throughout with truncated signatures and write $S^{(\leq m)}(X)_{a,b}$ when emphasising the truncation level. The total number of entries up to level m on a d -letter alphabet is, for $d > 1$,

$$\#\{\text{entries up to level } m\} = \sum_{k=0}^m d^k = \frac{d^{m+1} - d}{d - 1}. \quad (3)$$

For a single line segment $X(t) = (\alpha_1 + \beta_1 t, \dots, \alpha_d + \beta_d t)$, $t \in [a, b]$, the iterated integrals are explicit at all levels:

$$S(X)_{a,b}^{i_1, \dots, i_k} = \frac{\beta_{i_1} \beta_{i_2} \dots \beta_{i_k} (b - a)^k}{k!}. \quad (4)$$

This formula makes line segments the natural building blocks for the inversion of Section 4.

The Mathematica function `signat[paths, limits, level]` that computes (2) symbolically is in the companion notebook `CoreSignatures.nb`. Worked signatures of the cubic $\{t, t^3\}$, the unit circle $\{\cos t, \sin t\}$, and the family $\{t, \sin(k\pi t^q)\}$ for $k = 1, \dots, 3$ and $q = 1, 2$ — including the closed forms in terms of the Fresnel integrals — are tabulated in the same notebook.

2.2 Properties

2.2.1 Translation invariance

The signature (4) depends only on the slopes β_i and the duration $b - a$, not on the offsets α_i . This holds in general: the signature encodes the shape of the path modulo translation, so the initial value X_a must be tracked separately when needed.

2.2.2 Shuffle product

A (p, q) -shuffle of two ordered sequences of lengths p and q is an interleaving that preserves the relative order within each sequence. Writing the shuffle of two multi-indices I, J as $I \sqcup J$ (the multiset of all valid interleavings), the *shuffle product identity* for any path X states

$$S(X)_{a,b}^I \cdot S(X)_{a,b}^J = \sum_{K \in I \sqcup J} S(X)_{a,b}^K. \quad (5)$$

Equation (5) is the algebraic engine of Section 3: it provides the polynomial relations among signature entries that allow us to identify a free generating set. The recursive implementation of $\sqcup$ used in the companion notebook is the top answer of [this Mathematica.SE thread](#).

2.2.3 Chen's identity

Given consecutive paths $X : [a, b] \rightarrow \mathbb{R}^d$ and $Y : [b, c] \rightarrow \mathbb{R}^d$, the concatenation $Z = X * Y : [a, c] \rightarrow \mathbb{R}^d$ has signature given by Chen's identity ([Lyons, Caruana and Lévy (2007)]):

$$S(Z)_{a,c} = S(X)_{a,b} \otimes S(Y)_{b,c}. \quad (6)$$

Componentwise, abbreviating $x_w = S(X)_{a,b}^w$, $y_w = S(Y)_{b,c}^w$, $z_w = S(Z)_{a,c}^w$, with the conventions $x_\emptyset = y_\emptyset = 1$,

$$z_{i_1, \dots, i_k} = \sum_{j=0}^k x_{i_1, \dots, i_j} \cdot y_{i_{j+1}, \dots, i_k}. \quad (7)$$

The convolution form (7) lets us work with individual entries of Z without reconstructing the full concatenated signature — this is essential for the inversion algorithm of Section 4.

3 Core Signatures

The shuffle relations (5) mean that not every entry of the truncated signature carries new information. The classical remedy is the log signature, of dimension equal to that of the free nilpotent Lie algebra $\mathfrak{g}^m(\mathbb{R}^d)$. We show that a subset of the signature entries themselves — the *core elements* — has the same cardinality, can be read directly from $S(X)_{a,b}$, and admits an explicit structural characterisation in terms of Lyndon words.

3.1 Core variables via the shuffle product

Let $V_m = \{s_w : |w| = m\}$ denote the set of signature entries at level m , and let $C_m \subseteq V_1 \cup \dots \cup V_m$ be the *core set up to level m* , defined inductively as the entries that cannot be expressed in terms of lower-indexed and lower-level entries via the shuffle relations. We compute C_m for $d = 2$ by recursively applying (5) to all pairs (I, J) with $|I| + |J| = m$, treating the resulting linear system in V_m with coefficients in $\mathbb{R}[V_1 \cup \dots \cup V_{m-1}]$.

3.1.1 Worked example: $d = 2$, level 3

The shuffle product of one-letter and two-letter indices on $\{1, 2\}$ generates eight equations among the eight entries of $V_3 = \{s_{1,1,1}, s_{1,1,2}, \dots, s_{2,2,2}\}$. After solving in terms of $V_1 \cup V_2$ and the previously identified $C_2 = \{s_1, s_2, s_{1,2}\}$, two entries remain free; we choose the lex-minimal representatives in each shuffle orbit:

$$C_3 = C_2 \cup \{s_{1,1,2}, s_{1,2,2}\} = \{s_1, s_2, s_{1,2}, s_{1,1,2}, s_{1,2,2}\}.$$

The reduced Gröbner basis of the ideal of shuffle relations at level 3 is

$$\begin{aligned} & 3s_{1,1,1} - s_1s_{1,1}, \\ & -s_2s_{1,1} + s_1s_{2,1} + s_{1,1,2} - s_{2,1,1}, \\ & -s_1s_{2,1} + s_{1,2,1} + 2s_{2,1,1}, \\ & -s_2s_{1,2} + s_2s_{2,1} + 2s_{1,2,2} - 2s_{2,2,1}, \\ & -s_2s_{2,1} + s_{2,1,2} + 2s_{2,2,1}, \\ & 3s_{2,2,2} - s_2s_{2,2}. \end{aligned} \tag{8}$$

Each line groups together one shuffle orbit and pins down the substitution rules

$$\{s_{1,1,1}, s_{1,2,1}, s_{2,1,1}, s_{2,1,2}, s_{2,2,1}, s_{2,2,2}\} \mapsto \text{polynomials in } C_3,$$

listed in Appendix A as $Q_3 \subset Q_5$.

3.1.2 Levels 4 to 6 in summary

The same computation at higher levels yields the core sizes

$$|C_m| \text{ for } m = 1, \dots, 6 = 2, 3, 5, 8, 14, 23, \tag{9}$$

matching the log-signature column of [Reizenstein (2015)], Table 2 (reproduced as Table 1). For $d = 3$ we obtain 3, 6, 14, 32, 80, 196, again matching the log-signature size. The Lyndon multi-indices that constitute C_4, C_5, C_6 for $d = 2$ are listed in Appendix A; the substitution rules Q_4, Q_5 are tabulated there in full and Q_6 for $d \in \{2, 3\}$ is exported in the companion notebook.

m	$d = 2$ sig	$d = 2$ core	$d = 2$ log	$d = 3$ sig	$d = 3$ log	$d = 4$ sig	$d = 4$ log
1	2	2	2	3	3	4	4
2	6	3	3	12	6	20	10
3	14	5	5	39	14	84	30
4	30	8	8	120	32	340	90
5	62	14	14	363	80	1364	294
6	126	23	23	1092	196	5460	964

Table 1: Sizes of signatures, core signatures, and log signatures at low levels. Core and log columns coincide; full table to $m = 12$ in [Reizenstein (2015)].

3.2 Core elements as Lyndon coordinates of $G^n(\mathbb{R}^d)$

The numerical coincidence $|C_m| = \dim \mathfrak{g}^m(\mathbb{R}^d)$ and the choice of the lex-minimal representative in each shuffle orbit are not coincidences. They are consequences of the Lyndon basis of the free Lie algebra.

3.2.1 Tensor algebra, shuffle Hopf algebra, and the universal variety

Let $V = \mathbb{R}^d$ and let $T(V) = \bigoplus_{n \geq 0} V^{\otimes n}$ be its tensor algebra, with basis the words on $\mathcal{A} = \{1, \dots, d\}$. The completion $T((V)) = \prod_{n \geq 0} V^{\otimes n}$ houses the formal series in which signatures live. The vector space $T(V)$ carries two compatible Hopf algebra structures ([Reutenauer (1993)] Sections 1.4–1.6):

- the *concatenation–desshuffle* structure $T(V)_\bullet$, with product $\otimes$ (juxtaposition of words) and coproduct $\Delta_\sqcup$ defined on letters by $\Delta_\sqcup(a) = a \otimes 1 + 1 \otimes a$ and extended as an algebra homomorphism for $\otimes$;
- the *shuffle–deconcatenation* structure $T(V)_\sqcup$, with the shuffle product $\sqcup$ of (5) and the deconcatenation coproduct $\Delta_\bullet(w) = \sum_{w=uv} u \otimes v$.

These two Hopf algebras are graded-dual under the canonical pairing $\langle u, v \rangle = \delta_{u,v}$ on basis words.

A series $g \in T((V))_\bullet$ is *group-like* if its empty-word coefficient is 1 and $\Delta_\sqcup(g) = g \otimes g$. The signature $S(X)_{a,b}$ of any bounded-variation path is group-like: writing $\sigma_w[X] = \langle S(X), w \rangle$, the shuffle product identity $\sigma_I \cdot \sigma_J = \sum_{K \in I \sqcup J} \sigma_K$ of (5) is exactly the dual statement that $\Delta_\sqcup(S(X)) = S(X) \otimes S(X)$. Equivalently, the linear functional $\chi_{S(X)} : T(V)_\sqcup \rightarrow \mathbb{R}$ defined by $w \mapsto \sigma_w[X]$ is an algebra homomorphism for the shuffle product — i.e., a *character* of the shuffle Hopf algebra.

Truncated at level n , the image

$$G^n(\mathbb{R}^d) := \{S^{(\leq n)}(X)_{a,b} : X \text{ a path of bounded variation in } \mathbb{R}^d\} \subset T^n(V)$$

is the step- n free nilpotent Lie group on d generators, an algebraic variety cut out exactly by the truncated shuffle relations. [Amendola, Friz and Sturmfels (2019)] call this the *universal variety* and identify its coordinate ring with the quotient of the polynomial algebra in the signature entries by the truncated shuffle ideal.

3.2.2 Lyndon words and the Lyndon basis theorem

Fix the lexicographic order on $\mathcal{A} = \{1, \dots, d\}$. A non-empty word w on $\mathcal{A}$ is a *Lyndon word* if $w < vu$ in lexicographic order for every non-trivial factorisation $w = uv$ with u, v non-empty. For $d = 2$, the Lyndon words of length ≤ 6 on $\{1, 2\}$ are listed in Table 2; comparison with C_6 as listed in Appendix A shows the multi-indices agree exactly.

Length	Lyndon words on $\{1, 2\}$
1	1, 2
2	12
3	112, 122
4	1112, 1122, 1222
5	11112, 11122, 11212, 11222, 12122, 12222
6	111112, 111122, 111212, 111222, 112122, 112212, 112222, 121222, 122222

Table 2: Lyndon words on $\{1, 2\}$ up to length 6. These are the multi-indices of the core variables.

Theorem 1 (Lyndon basis of the free Lie algebra). *The Lyndon brackets $\{[w] : w \text{ Lyndon on } \mathcal{A}\}$ form a homogeneous basis of the free Lie algebra $\mathfrak{L}(\mathbb{R}^d)$, where the bracket is defined recursively by $[a] = a$ for letters and $[w] = [[u], [v]]$ for the canonical Lyndon factorisation $w = uv$ with v the longest Lyndon proper suffix of w .*

This is Reutenauer’s Theorem 5.1 ([Reutenauer (1993)]); the version restated for the universal signature variety appears as [Amendola, Friz and Sturmfels (2019)] Proposition 4.7.

3.2.3 From PBW to Lyndon coordinates

The Lyndon basis theorem becomes a statement about coordinates on $G^n(\mathbb{R}^d)$ via three classical inputs.

(i) **Poincaré–Birkhoff–Witt.** The universal enveloping algebra $U(\mathfrak{L}(V))$ of the free Lie algebra is canonically isomorphic to the concatenation–deshuffle Hopf algebra $T(V)_\bullet$, with the inclusion $\mathfrak{L}(V) \hookrightarrow T(V)_\bullet$ realised by the bracket $[a, b] = a \otimes b - b \otimes a$ on letters ([Reutenauer (1993)] Theorem 0.5). Combined with Theorem 1, PBW gives a vector-space basis of $T(V)_\bullet$ consisting of ordered concatenated monomials in the Lyndon brackets:

$$\left\{ [w_1]^{\otimes n_1} \otimes [w_2]^{\otimes n_2} \otimes \dots \otimes [w_k]^{\otimes n_k} : w_1 < \dots < w_k \text{ Lyndon, } n_i \geq 1 \right\}.$$

(ii) **Radford’s theorem on the shuffle algebra.** Dualising the PBW basis under the canonical pairing $\langle u, v \rangle = \delta_{u,v}$ transports it to a basis of $T(V)_\sqcup$ given by shuffle monomials in the Lyndon words themselves.

Theorem 2 (Radford 1979). *The shuffle algebra $(T(V), \sqcup)$ is the free commutative algebra on the set $\mathcal{L}$ of Lyndon words on $\mathcal{A}$:*

$$(T(V), \sqcup) \cong \mathbb{R}[w : w \in \mathcal{L}].$$

Every element of $T(V)$ has a unique expression as a polynomial in the Lyndon words under the shuffle product.

For a proof see [Reutenauer (1993)] Theorem 6.1.

(iii) Group-likes as characters. By the discussion of Section 3.2.1, every group-like element $g \in T((V))$ corresponds bijectively to an algebra homomorphism (a character) $\chi_g : (T(V), \sqcup) \rightarrow \mathbb{R}$ via $\chi_g(w) = \langle g, w \rangle$. By Theorem 2, a character of a polynomial algebra is determined by its values on the polynomial generators. Hence the coordinate map

$$G^\infty(\mathbb{R}^d) \xrightarrow{\sim} \mathbb{R}^{\mathcal{L}}, \quad g \mapsto (\sigma_w[g])_{w \in \mathcal{L}},$$

is a bijection: the Lyndon-indexed signature entries are algebraically independent coordinates on the universal variety, and every other entry σ_v (v non-Lyndon) is the unique polynomial P_v in Lyndon coordinates of length $\leq |v|$ obtained by writing v as a shuffle monomial in Lyndon factors. Truncating at level n gives algebraically independent coordinates on $G^n(\mathbb{R}^d)$ indexed by Lyndon words of length $\leq n$.

(iv) Lex-minimal representatives are Lyndon. The remaining ingredient ties the choice made by the shuffle-product elimination of Section 3.1 to the Lyndon representatives:

Lemma 3 (Duval). *Every word w on $\mathcal{A}$ admits a unique factorisation $w = u_1 u_2 \cdots u_k$ with $u_1 \geq u_2 \geq \cdots \geq u_k$ all Lyndon. Equivalently, a primitive necklace — the cyclic equivalence class of a primitive word — has a unique lex-minimal rotation, and that rotation is the Lyndon representative of the necklace.*

See [Reutenauer (1993)] Theorem 5.1.5 for the proof.

3.2.4 Identification of core variables with Lyndon coordinates

We now combine the three classical ingredients of Section 3.2.3 with the elimination computation of Section 3.1 into two separate identifications. The first — structural — uses only Radford's theorem and is independent of any elimination order:

Proposition 4 (Structural: Lyndon coordinates as a transcendence basis). *For each $m \geq 1$, the signature entries indexed by Lyndon words of length $\leq m$ form a transcendence basis for the coordinate ring of the universal variety:*

$$\mathcal{O}(G^m(\mathbb{R}^d)) = \mathbb{R}[\sigma_w : |w| \leq m] / \mathcal{I}_{\sqcup}^{(\leq m)} \cong \mathbb{R}[\sigma_w : w \in \mathcal{L}, |w| \leq m]. \quad (10)$$

Every σ_v with v non-Lyndon and $|v| \leq m$ is the unique polynomial P_v in the Lyndon entries of length $\leq |v|$ determined by writing v as a shuffle monomial in Lyndon factors.

Proof. By Theorem 2, the shuffle algebra $(T(V), \sqcup)$ is the polynomial algebra on the set $\mathcal{L}$ of Lyndon words. Truncating at level m and quotienting by the shuffle relations of total degree $\leq m$ yields (10). The uniqueness of P_v follows from the polynomial-algebra structure of the right-hand side. $\square$

The second — computational — identifies the specific representatives that the elimination procedure of Section 3.1 picks out, using the lex-minimal $\leftrightarrow$ Lyndon correspondence of Lemma 3:

Proposition 5 (Computational: lex-minimal elimination selects Lyndon representatives). *The shuffle-product elimination of Section 3.1, applied with the lex-minimal-representative-first ordering used by the Mathematica solver, retains as free variables at level n exactly the entries $\{\sigma_w : w \in \mathcal{L}, |w| = n\}$ indexed by Lyndon words of that length.*

Proof. At level n , the shuffle relations $\sigma_I \sqcup \sigma_J = \sum_{K \in I \sqcup J} \sigma_K$ with $|I| + |J| = n$ form a linear system in $V_n = \{\sigma_w : |w| = n\}$ with coefficients in $\mathbb{R}[V_1 \cup \dots \cup V_{n-1}]$. By the Lyndon basis count of Theorem 1, the kernel of this linear system has rank $\ell_d(n)$, the number of Lyndon words of length n . Eliminating any system of representatives — one per cyclic equivalence class of length- n multi-indices — produces a complete set of $\ell_d(n)$ free variables.

The lex-minimal-first ordering used by Mathematica’s `Solve` prioritises the elimination of higher-indexed variables, so within each cyclic class it retains the lex-minimal element. By Lemma 3, the lex-minimal element of a primitive cyclic class is the Lyndon word of that class. Non-primitive classes — those of words u^k with u primitive and $|u| < n$ — are entirely expressible in lower-level Lyndon coordinates via the shuffle-power identity $\sigma_u^{\sqcup k} = k! \sigma_{u^k} + (\text{terms in lower-level } \sigma)$, and are therefore eliminated at this step using free variables from earlier levels.

Hence the $\ell_d(n)$ free variables retained at level n are exactly the Lyndon-indexed entries of that length. $\square$

Combining the two propositions recovers the result that motivated this section:

Corollary 6 (Core variables are Lyndon coordinates). *The core sets C_m identified by the shuffle-product computation of Section 3.1 satisfy*

$$C_m = \{\sigma_w : w \in \mathcal{L}, |w| \leq m\},$$

and the substitution rules Q_m tabulated in Appendix A are the explicit form of the truncated Radford isomorphism (10) of Proposition 4.

Proof. Proposition 5 identifies C_m with the Lyndon-indexed entries of length $\leq m$; Proposition 4 identifies those entries with the transcendence basis of the coordinate ring via the truncated Radford isomorphism. The substitution rules Q_m make the inverse isomorphism explicit on chosen representatives. $\square$

3.3 Witt formula and the Pascal-triangle structure

The number of Lyndon words of length n on a d -letter alphabet is given by Witt’s formula ([Reutenauer (1993)] Section 4):

$$\ell_d(n) = \frac{1}{n} \sum_{k|n} \mu\left(\frac{n}{k}\right) d^k, \quad (11)$$

where μ is the Möbius function. Refining by the composition $(n_1, \dots, n_d)$ of letter multiplicities,

$$\ell_d(n_1, \dots, n_d) = \frac{1}{n} \sum_{k \mid \gcd(n_1, \dots, n_d)} \mu(k) \frac{(n/k)!}{(n_1/k)! \cdots (n_d/k)!}. \quad (12)$$

Corollary 7 (Size of the core sets). *For $d = 2$ and any $m \geq 1$, the number of core variables of length $n \leq m$ with exactly j occurrences of the letter 1 is $\ell_2(j, n - j)$, and $|C_m| = \sum_{n=1}^m \ell_2(n)$.*

The approximation

$$\text{core2app}(m, k) = \left\lfloor \frac{1}{m} \binom{m}{k} \right\rfloor, \quad (13)$$

empirically observed in the 2023 version of this paper, is exact whenever $\gcd(k, m - k) = 1$ and underestimates the count by exactly the number of necklaces of non-trivial period otherwise; the first divergence occurs at $m = 10$. The Pascal-triangle structure of Table 3 follows directly from (12).

$m \backslash j$	5	4	3	2	1	0
1					1	1
2					1	
3				1	1	
4			1	1	1	
5		1	2	2	1	
6	1	2	3	2	1	

Table 3: Number of core variables ($d=2$) of length m with exactly j occurrences of the letter 1, computed from Witt’s formula (12).

3.4 The Gröbner reconstruction as a constructive corollary

The substitution rules Q_m tabulated in Appendix A exhibit the polynomial isomorphism

$$\mathbb{R}[\sigma_w : w \text{ Lyndon}, |w| \leq m] \xrightarrow{\sim} \mathbb{R}[\sigma_w : |w| \leq m] / \mathcal{I}_{\text{shuffle}}^{(m)}$$

explicitly. The advantage of computing Q_m via the shuffle Gröbner basis, rather than invoking Theorem 1 directly, is that it returns the inverse map in closed form: given a target core signature, the rules Q_m recover the full truncated signature $S^{(\leq m)}(X)$ as a polynomial in the core entries. This explicit inverse is what we will use in Section 4 to invert the signature into a piecewise linear path.

4 Inversion of Core Signatures

We use Chen’s identity (7) to recover, from a target core signature, a piecewise linear path whose own core signature exactly matches the target up to a chosen truncation level. We first treat the time-ordered case $X(t) = (t, f(t))$, where the algebra closes after two or three segments, and then turn to the general two-dimensional case and to the multi-segment regime.

Related work on the inversion problem. The inversion problem has been developed extensively from an algebraic-geometric perspective by the Berlin/TU group. [Amendola, Lotter and Schmitz (2026)] (Améndola-Lotter-Schmitz, 2026) generalise the piecewise linear and polynomial classes of [Amendola, Friz and Sturmfels (2019)] to piecewise polynomial paths and splines, and introduce the *path recovery degree* PRdeg (Definition 6.1 of that paper) as the number of complex points in a generic fibre of the signature map. This formalises what we observe empirically: the two-piece time-ordered system has two algebraic solutions, the three-piece system extends to a finite set of algebraically determined choices, and the multi-piece case admits a finite recovery degree that depends on the segment-degree composition. Their Table 3 tabulates PRdeg for all piecewise-polynomial classes with total degree 4 in $d = 2$ at level $k = 4$: piecewise linear with four segments yields PRdeg = 4, higher-degree spline classes go up to PRdeg = 40. [Schmitz (2025)] (Schmitz, 2025) gives an $O(d^4)$ algorithm for the path-recovery problem at truncation level 3 with d segments in d dimensions, using Gauss transformations on signature tensors as a tensor analogue of congruence normal forms, and demonstrates that this approach outperforms Gröbner-basis methods at higher d . [Amendola and Schmitz (2025)] (Améndola-Schmitz, 2025) treats the adjacent problem of recovering a path from a *barycenter* of multiple signatures, relevant when the input is an expected signature of a stochastic process rather than the signature of a single path.

Our approach is elementary in technique and complementary in framing. The Chen-identity machinery applies to any piecewise linear path in $\mathbb{R}^d$; we develop the time-ordered case $X(t) = (t, f(t))$ in detail because it is the case most relevant for term-structure modelling and because the algebra closes with closed-form expressions for the two-segment and three-segment systems (Appendix B). The same algorithm extends to general two-dimensional paths — see Section 4.4 and the semicircle and digit-5 examples therein — and in principle to arbitrary dimension, with the per-segment parametrisation $\{(a_j t, b_j t)\}$ replaced by the appropriate d -dimensional one. For multi-piece inversion we use a weighted regression in signature space (Section 4.5) and develop a moment-matching dictionary that links signature-space minimisation to classical L^1 approximation theory via the Lyndon coordinates of Section 3.2; that link is, to our knowledge, not present in the algebraic-geometric line of work. The path-recovery-degree count of [Amendola, Lotter and Schmitz (2026)] would, in our setting, equal the number of algebraically equivalent inversions obtainable by choosing which equation to drop in the overdetermined Chen-identity system; the regression of Section 4.5 provides a numerically stable selection criterion among these algebraic alternatives.

4.1 Time-ordered paths: two pieces

Let $X(t) = (t, f(t))$ on $[0, T]$ and assume the target core signature $\{z_1, z_2, z_{1,2}, z_{1,1,2}, z_{1,2,2}\}$ comes from a smooth f , away from the adversarial cases discussed in Section 4.2. We seek the concatenation $X * Y$ of two line segments

$$X(t)|_{[0,t_1]} = (t, \beta_1 t), \quad Y(t)|_{[t_1,T]} = (t, \beta_1 T + \beta_2 t),$$

parametrised by $\{T, t_1, \beta_1, \beta_2\}$, whose core signature matches the target. By (4) and (7), the matching conditions are the five equations of Table 4.

Level	Eq #	Equation
1	1	$T = z_1$
1	2	$\beta_1 t_1 + \beta_2 (T - t_1) = z_2$
2	3	$\frac{1}{2}\beta_1 t_1^2 + \frac{1}{2}\beta_2 (T - t_1)^2 + \beta_2 t_1 (T - t_1) = z_{1,2}$
3	4	$\frac{1}{6}\beta_1 t_1^3 + \frac{1}{6}\beta_2 (T - t_1)^3 + \frac{1}{2}\beta_2 t_1 (T - t_1)^2 + \frac{1}{2}\beta_2 t_1^2 (T - t_1) = z_{1,1,2}$
3	5	$\frac{1}{6}\beta_1^2 t_1^3 + \frac{1}{6}\beta_2^2 (T - t_1)^3 + \frac{1}{2}\beta_2^2 t_1 (T - t_1)^2 + \frac{1}{2}\beta_1 \beta_2 t_1^2 (T - t_1) = z_{1,2,2}$

Table 4: System for the two-piece time-ordered inversion.

The system is overdetermined (5 equations, 4 unknowns) and admits two solutions, obtained by dropping equation #5 or equation #4; the closed-form expressions for $\{T, t_1, \beta_1, \beta_2\}$ in each case are listed in Appendix B. Substituting either solution into the dropped equation gives the same consistency condition

$$2z_{1,2}^2 + z_1 z_2 z_{1,2} = 3z_1 z_{1,2,2} + 3z_2 z_{1,1,2}, \quad (14)$$

which expresses the algebraic constraint imposed by the limited freedom of two segments.

Worked examples

For $f(t) = \sin(\pi t)$ on $[0, 1]$ the target core signature is $\{1, 0, -2/\pi, -1/\pi, 1/4\}$. Solution 1 (dropping equation #5) yields $\{T, t_1, \beta_1, \beta_2\} = \{1, 1/2, 8/\pi, -8/\pi\}$, giving the symmetric tent with peak $(1/2, 4/\pi)$. Solution 2 fails because $z_2 = 0$. For $f(t) = \sin(\pi t^2)$ Solution 1 gives the asymmetric tent with peak slightly past $t = 1/2$, matching the geometric asymmetry of the curve (Figure 1). For $f(t) = \sin(\pi t^2) + 1 - t$ both solutions exist and produce two valid two-segment fits whose crossing point lies very close to the original curve (Figure 2); the two values of t_1 bracket the location $t \approx 0.832$ of maximum $|f''|$.

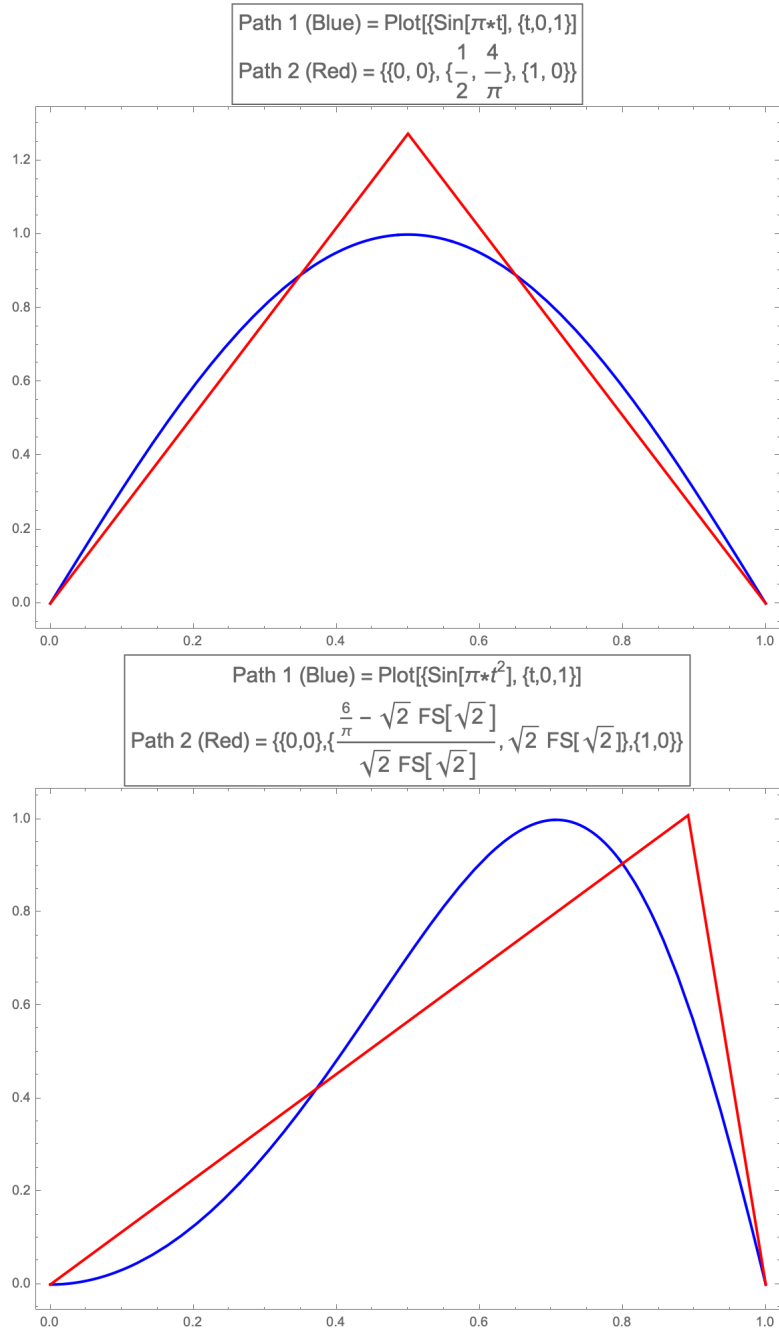


Figure 1: Two-segment fits to $\{t, \sin(\pi t)\}$ and $\{t, \sin(\pi t^2)\}$.

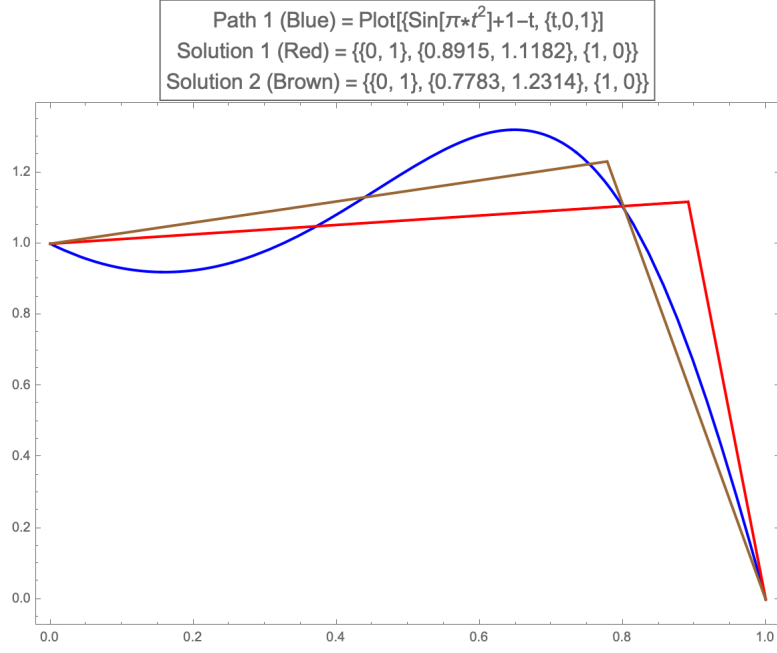


Figure 2: The two two-segment fits to $\{t, \sin(\pi t^2) + 1 - t\}$.

For an additive target $\{t, \alpha t^a + \beta t^b\}$ with $a, b > 0$, the core signature decomposes additively in $z_1, z_2, z_{1,2}, z_{1,1,2}$ across the two summand paths and adds a single cross-term in $z_{1,2,2}$. This explains why the choice of which equation to drop affects the fit qualitatively: for paths well-modelled as a primary component plus a linear drift, dropping equation #4 recovers the primary component more faithfully.

4.2 Adversarial cases

The two-piece inversion fails when the closed forms of Appendix B have vanishing denominators. The two relevant degeneracies are:

- $z_1 z_2 - 2z_{1,2} = 0$ (Lévy area zero relative to the chord), where the intermediate point's distance from the chord is undetermined;
- $z_2 = 0$, which removes one of the two solutions.

Both situations occur for $\{t, \sin(2\pi t)\}$ and similar oscillatory targets where one period of the oscillation cancels another. The remedy is to use three or more pieces.

4.3 Three pieces and beyond

Adding a third segment introduces three new unknowns and two more core entries (level 4) into the system. The resulting equations are listed in Appendix B. For four or five segments we move to level 5 of the signature; the choice of which equation to drop in the resulting overdetermined system is no longer canonical, and the non-linearity of the equations makes symbolic solution prohibitively slow. We solve numerically through the regression of Section 4.5.

Figures 3 and 4 show four- and five-segment fits to $\{t, \sin(k\pi t)\}$ and $\{t, \sin(k\pi t^2)\}$ for $k = 1, 2, 3$. The recovery is exact at the algebraic level for piecewise linear targets and degrades gracefully for oscillatory smooth targets, in line with the per-segment-frequency analysis of Section 4.5.

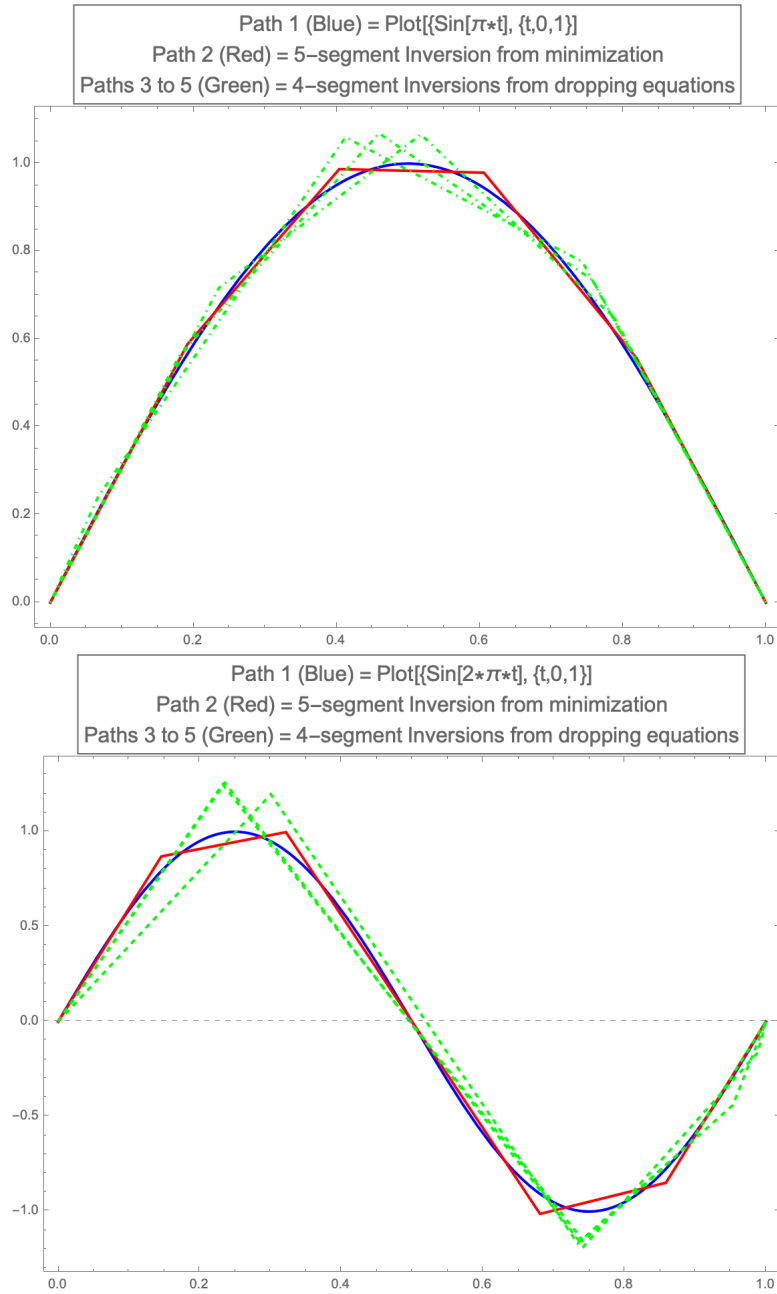


Figure 3: Fits of 4 and 5 segments to $\{t, \sin(k\pi t)\}$.

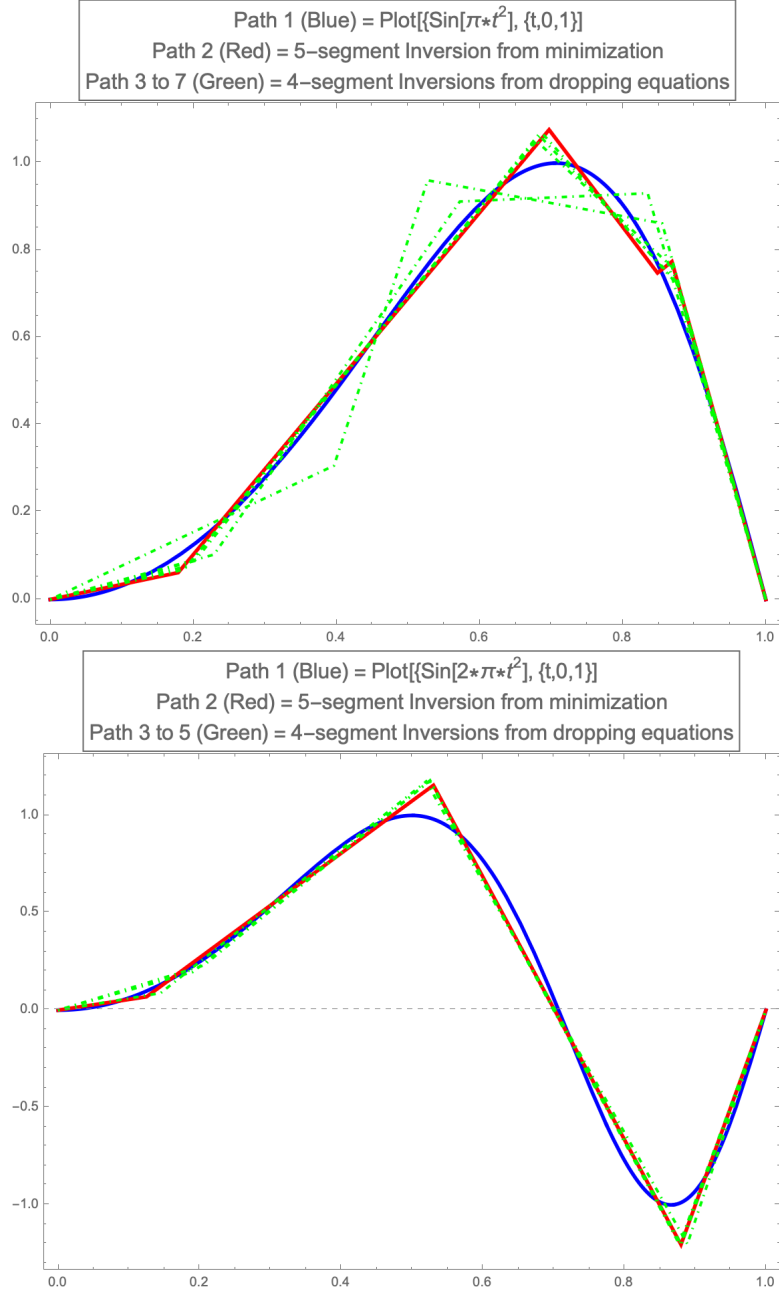


Figure 4: Fits of 4 and 5 segments to $\{t, \sin(k\pi t^2)\}$.

4.4 General 2d paths

For a general path $X(t) = (f(t), g(t))$, both coordinates carry information and the path may loop. We replace each segment by $\{a_j t, b_j t\}$ and apply the same Chen-based matching, filtering out reconstructed paths whose total length exceeds an empirical threshold (chosen by inspection of the algebraic-extraneous solutions in each test case). *This length-based filter is heuristic, not algebraic: it discards solutions of the Chen system whose excess arclength signals a spurious algebraic root rather than a geometric reconstruction. A principled replacement — e.g. a constraint derived from the path recovery degree of [Amendola, Lotter and Schmitz (2026)] or from the regression of Section 4.5 —*

is desirable and is left to future work.

For the semicircle $\{\cos(\pi t), \sin(\pi t)\}$ with $t \in [0, 1]$ the three-piece fit recovers the visual outline cleanly (Figure 5). For the digit “5” described by the eight control points

$$(0, 20, 64, 79, 35, 37, 71, 100; 0, -3, 13, 47, 54, 77, 96, 93),$$

both the three-piece and five-piece inversions reproduce the topology of the digit (Figures 6 and 7); the five-piece fit, computed via `NMinimize` with random search, matches the original 7-segment piecewise linear path to visual accuracy.

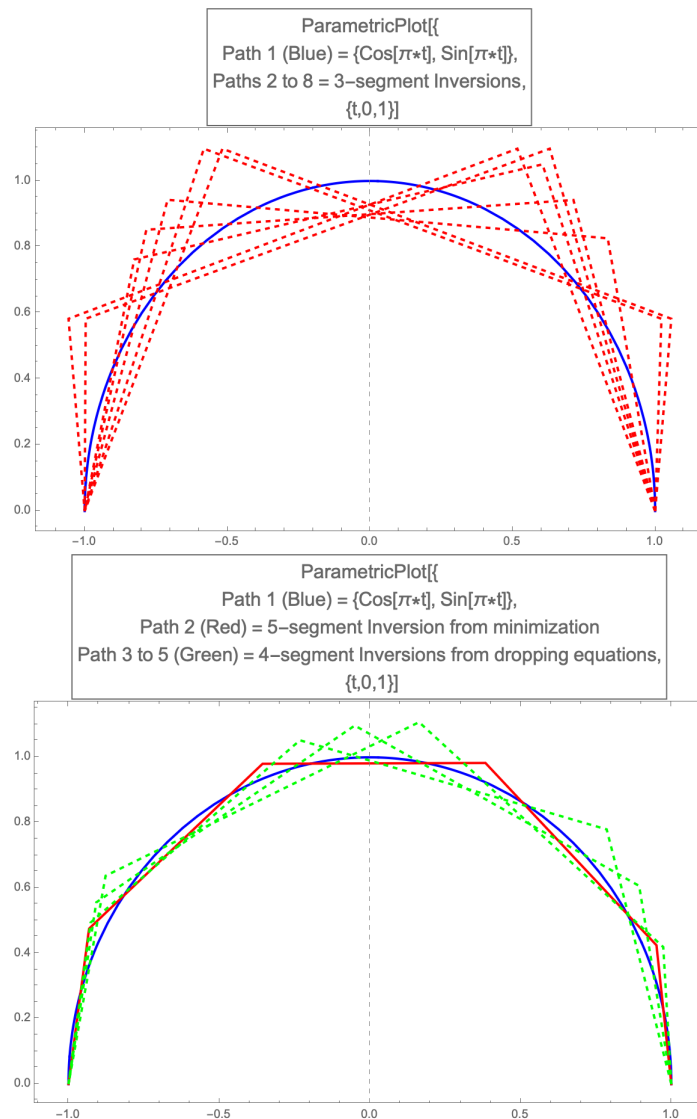


Figure 5: Three- and four-piece inversions of the semicircle $\{\cos(\pi t), \sin(\pi t)\}$.

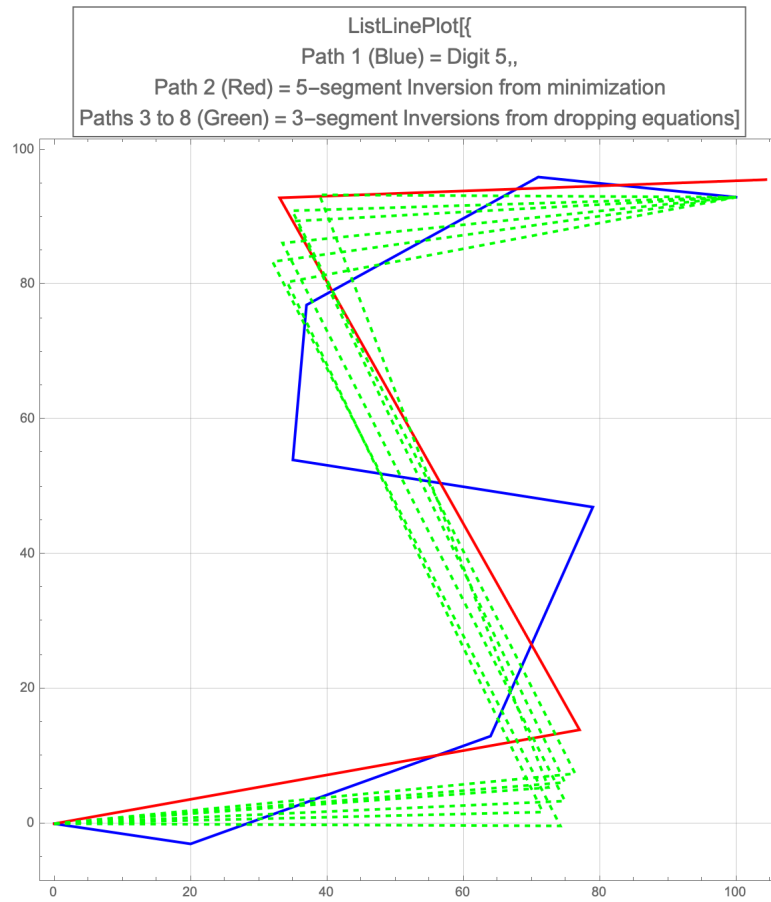


Figure 6: Three-segment inversion of the digit 5.

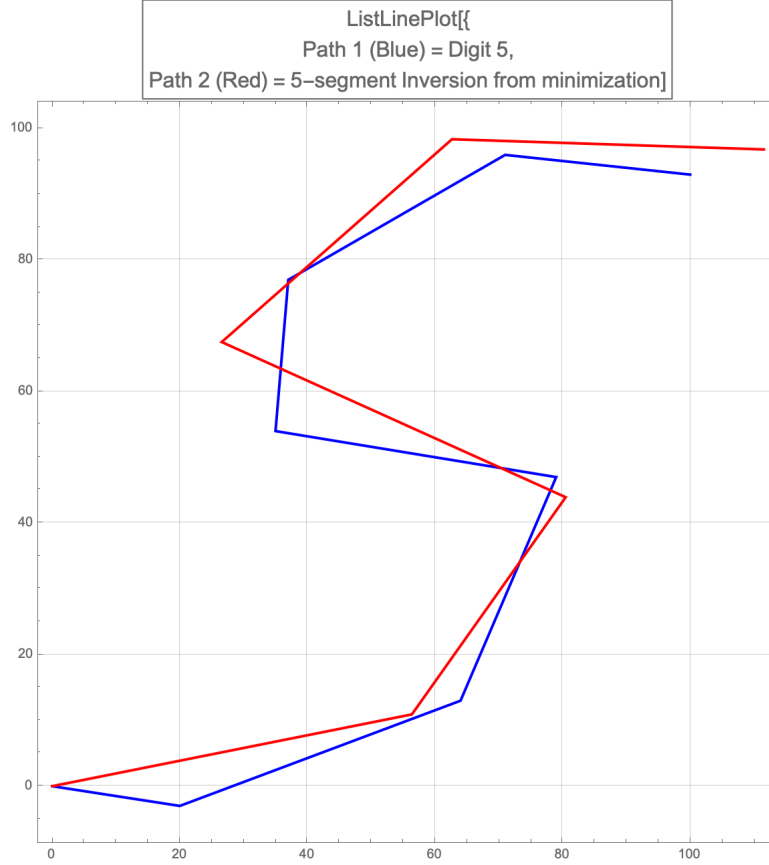


Figure 7: Five-segment inversion of the digit 5.

4.5 Inversion as weighted regression in signature space

For piecewise linear paths with n segments the inversion problem has $2n - 2$ free parameters once the endpoints are pinned, exactly matching the number of independent core entries up to level n from Corollary 7. To use signature information beyond level n — which the per-segment-frequency analysis below shows is necessary for oscillatory targets — the system becomes overdetermined, and we minimise a weighted least-squares objective in signature space:

$$\theta^* = \arg \min_{\theta} \sum_{w \in \mathcal{L}_m} w_w (\sigma_w[g_\theta] - \sigma_w[X])^2, \quad (15)$$

where $\mathcal{L}_m$ is the set of Lyndon words up to level m . The companion notebook uses level-only weights $w_w = W_{|w|}$ with $W_\bullet = \{50, 50, 30, 10, 10, 3, 3, 3\}$ across the core indices and **DifferentialEvolution** as the optimiser. For targets within the parametric class $\mathcal{P}_n$ — i.e. signatures generated from a piecewise-linear path of the right segment count — the regression recovers the generating parameters to floating-point precision. For smooth out-of-class targets, the regression converges to a weighted least-squares optimum that differs from the closed-form algebraic solutions of Appendix B and achieves a strictly lower $J(\theta^*)$ than any of them; the gap quantifies how much cross-residual budget the regression is exploiting. The practical implementation choices are summarised in Algorithm 1.

Algorithm 1 Core-signature inversion via signature-space regression

Inputs. Target core signature $\{\sigma_w[X] : w \in \mathcal{L}_m\}$; number of segments n ; truncation level $m \geq n$; level-weight schedule $W = (W_1, \dots, W_m)$.

Output. Piecewise-linear path $g_{\theta^*} \in \mathcal{P}_n$ whose core signature matches $\sigma[X]$ in the weighted-least-squares sense of (15).

1. *Parametrise.* Write g_θ as a piecewise-linear path with n segments parametrised by $\theta = (\beta_1, \dots, \beta_n; t_1, \dots, t_{n-1})$, endpoints pinned, subject to the ordering constraint $0 < t_1 < \dots < t_{n-1} < T$.
2. *Compile the objective.* Compute the core signature $\sigma_w[g_\theta]$ symbolically using Chen’s identity (7) and the line-segment formula (4), for all $w \in \mathcal{L}_m$. This step is one-time per (n, m) pair and produces explicit polynomials in θ . Form $J(\theta) = \sum_{w \in \mathcal{L}_m} W_{|w|} (\sigma_w[g_\theta] - \sigma_w[X])^2$.
3. *Optimise.* Call `NMinimize[J[ $\theta$ ], constraints,  $\theta$ , Method  $\rightarrow$  DifferentialEvolution]`. Tolerances: `AccuracyGoal  $\rightarrow$  8`, `PrecisionGoal  $\rightarrow$  8`, `MaxIterations  $\rightarrow$  500`. The Mathematica defaults for `DifferentialEvolution` (population size $\approx 30 \dim(\theta)$, `CrossProbability  $\rightarrow$  0.5`, `ScalingFactor  $\rightarrow$  0.6`) suffice for $n \leq 4$ segments and $m \leq 6$ levels. Typical convergence on smooth out-of-class targets at level $m = 3$ truncation is to $J(\theta^*)$ of order 10^{-3} and improves with m ; on in-class targets the regression reaches floating-point precision.
4. *Diagnose and return.* Empirically, on the time-augmented targets tested in this paper at $n = 2$, DE converges to a single basin across independent `RandomSeed` values, and the spread of $J(\theta^*)$ across seeds serves as a useful convergence diagnostic. At higher segment counts the universal-variety path-recovery-degree formalism of [Amendola, Lotter and Schmitz (2026)] (Definition 6.1; their Table 3 records $\text{PRdeg} = 4$ for piecewise linear paths with four segments at $d = 2$, $k = 4$) implies that the regression objective may have multiple basins corresponding to algebraically distinct paths with the same level- m signature. Whether the weighted regression (15) retains a single dominant basin in such regimes is an open question; the cross-seed $J(\theta^*)$ spread is the appropriate empirical diagnostic. The algebraic Solution-#1 and Solution-#2 of Appendix B are *not* local minima of J : they solve specific determined sub-systems (drop equation #4 or #5) and have non-zero gradient with respect to the dropped equation’s residual. They are nevertheless useful as diagnostic anchors — $J(\theta^*)$ should be bounded above by $\min_k J(\text{Sol-}\#k)$, and a small gap indicates a well-conditioned regression while a substantial gap indicates the regression is exploiting the cross-residual budget heavily. When the user wants the specific Solution-# k for downstream reasons (smoothness, monotonicity, or a closed-form expression in the target signature), they should solve the corresponding determined sub-system of Appendix B directly rather than the regression. Return θ^* and g_{θ^*} .

Remark 8 (Comparison-baseline parameters). The empirical-coincidence claim of Section 4.5.2 below uses the companion notebook `Frechet.nb`, which runs an independent $(\mu + \lambda)$ -

evolution strategy on the mean- L^1 objective rather than the signature-space objective (15). Parameters: population size 250; Gaussian mutation with standard deviation $\sigma = 0.025$ on interior breakpoints; elite-plus-mutation selection with fresh random “islands” injected each generation; fixed 50 generations with no early-stopping criterion. The mean- L^1 algorithm is independent of Algorithm 1 and serves to establish that the signature-space minimiser coincides with the optimal L^1 piecewise-linear approximant on the same partition class.

Remark 9 (Why the level scaling matters). Signature entries at level k on a path of duration T scale as $O(T^k/k!)$ via the simplex of integration. Without compensating weights the objective is dominated by noise in the high-level entries and loses sensitivity to the dominant low-level structure. The empirical weights above are decreasing with level but slower than $1/k!$; an explicit theoretical choice $w_w \propto (k!)^2/T^{2k}$ would normalise residuals to comparable magnitudes across levels, and is the form suggested by the asymptotic analysis below.

4.5.1 Signature entries as iterated path moments

For $X(t) = (t, f(t))$ on $[0, T]$, integration by parts gives

$$\sigma_{1,2}[X] = Tf(T) - \int_0^T f(t) dt, \quad (16)$$

$$\sigma_{1,1,2}[X] = \frac{T^2}{2}f(T) - \int_0^T t f(t) dt, \quad (17)$$

and so on. Every Lyndon-word signature entry of a time-augmented path is a polynomial in the moments $M_k(f) = \int_0^T t^k f(t) dt$ and the boundary values of f . To leading order in the level grading, the regression (15) is therefore a weighted moment-matching problem,

$$J(\theta) \simeq \sum_{k=0}^{m-1} \tilde{w}_k (M_k(g_\theta) - M_k(f))^2, \quad (18)$$

with effective weights $\tilde{w}_k$ derived from the w_w through the integration-by-parts identities. By the classical Hausdorff moment problem and its quantitative refinements ([Talenti (1987)]), matching the first m moments of $f \in C^s([0, T])$ pins down f in L^1 to order $O(m^{-s})$.

4.5.2 Empirical coincidence with mean L^1 minimisation

The companion notebook `Frechet.nb` runs an evolutionary algorithm on the four-segment class minimising the mean pointwise distance

$$\text{meandist}(g_\theta, f) = \frac{1}{N+1} \sum_{i=0}^N |f(t_i) - g_\theta(t_i)|, \quad t_i = iT/N.$$

On the target $f(t) = \sin(2\pi t^2)$ with $T = 1$ and $N = 1000$, the algorithm converges robustly across independent seeds to a piecewise-linear approximation whose interior breakpoints agree to within a few percent with those found by the algebraic core-signature inversion, but *not* to a common optimum: the two algorithms minimise structurally different objectives — signature-space residual sum for the algebraic inversion, mean L^1

distance for the evolutionary algorithm — and converge to distinct nearby minima rather than to a shared root. Numerical verification (Python port of both algorithms, population 250 over 200 generations across 10 independent seeds against a multi-start algebraic Chen solver) recovers the algebraic root at machine precision and the mean- L^1 optimum at a $\sim 11\%$ lower mean- L^1 value than the algebraic root, confirming the two minima are genuinely distinct. This is the multi-segment analogue of the single-segment 4.7% obstruction noted in Remark 13 below.

Mean- L^1 as an evaluation protocol for inversion algorithms. The role of `Fréchet.nb` in this paper is not that of a second inversion algorithm whose output is meant to coincide numerically with the algebraic Chen root. It is the role of an *independent test of the inversion in path space*: a method that knows nothing about signatures, minimises a clearly interpretable path-space distance (mean L^1 is the simplest choice), and returns the optimal piecewise-linear approximant in that distance. We propose this as a clean evaluation protocol for any signature-based inversion algorithm: pick a path-space distance with a clear geometric meaning, minimise it directly by an algorithm that does not use the signature structure, and use the resulting approximant as a benchmark against which the signature-based inversion is assessed. The 11% gap reported above is the answer the protocol returns for the present algebraic Chen inversion on $f(t) = \sin(2\pi t^2)$ at $n = 4$. The same protocol applied to the weighted regression (15) would, for a given choice of weights, quantify how far the regression’s path-space optimum sits from the path-space best-in-class approximant. Calibrating the weights w_w so as to minimise that gap on a representative training set of paths is the empirical complement to a proof of Conjecture 11; it is left to a subsequent paper.

Remark 10 (On the name “Fréchet”). Because the two paths share the time coordinate, $\text{EuclideanDistance}((t, f), (t, g_\theta)) = |f - g_\theta|$, so the notebook objective is mean L^1 , not the Fréchet distance between curves. The relevant comparison is therefore between the signature-regression solution and the optimal piecewise-linear L^1 approximant on the same partition class.

A heuristic explanation for the coincidence is now visible from (18): both methods determine the breakpoints by first-order conditions on identical leading moments of f , differing only by terms of order $\sup |f''| \cdot (\Delta t)^2$. We emphasise that this is an approximation-theoretic intuition rather than a proof of identity or norm equivalence; the conjectural status of the link is the subject of Conjecture 11 below.

4.5.3 The norm-equivalence conjecture

Conjecture 11 (Signature-norm equivalence). *Let $\mathcal{F} \subseteq C^1([0, T]; \mathbb{R})$ be a class of paths of bounded derivative and $X = (t, f)$, $G = (t, g)$ for $f, g \in \mathcal{F}$. There exist weights $\{w_w\}_{w \in \mathcal{L}}$ depending only on $|w|$ and T , with $w_w \sim (k!)^2/T^{2k}$ for $|w| = k$, and a Sobolev exponent $s \geq 0$ such that*

$$\sum_{w \in \mathcal{L}} w_w (\sigma_w[G] - \sigma_w[X])^2 \asymp \|f - g\|_{H^s([0, T])}^2 \quad (19)$$

uniformly for f, g in compact subsets of $\mathcal{F}$, with constants of equivalence depending only on the diameter of the compact set in H^s .

Remark 12 (Status). We have: (i) the empirical coincidence on a finite test set; (ii) the moment-matching reason of (18); (iii) the classical link between moment matching and L^1/L^2 approximation. We do not have: an explicit identification of s , the constants of equivalence, or a proof for any subclass of $\mathcal{F}$ richer than polynomials. The natural first-proof target is therefore the class of *polynomial paths* on $[0, T]$: the universal-variety coordinates restricted to polynomial paths are described explicitly in [Amendola, Friz and Sturmfels (2019)] Section 5 and in [Pfeffer, Seigal and Sturmfels (2018)], and the moment-matching identities of (16) reduce there to finite-dimensional linear systems in the polynomial coefficients, so a proof of (19) at that level would not require new functional-analytic machinery. A positive answer would identify the inversion of this section as a Galerkin projection of f onto $\mathcal{P}_n$ in a Sobolev norm, with the level m playing the role of the truncation parameter.

An empirical complement to a proof is to calibrate the weights w_w on representative training sets — by genetic search, gradient descent through the inner least-squares, or otherwise — and to test whether the calibrated asymptotic matches the $(k!)^2/T^{2k}$ worst-case form (expected for rough paths where level- k entries can saturate the $O(T^k/k!)$ bound and one wants each level to contribute comparably to the objective), the empirically observed decreasing form (expected for smooth targets where the actual entries shrink faster than the worst-case bound, so signal-to-noise decreases with level and the high-level weights should be small), or something between. Training across *both* smoothness regimes — e.g., rough paths constructed from fractional Brownian motion alongside smooth analytic targets — would identify the class on which the conjecture holds in the simple $(k!)^2/T^{2k}$ form, the class on which the empirical decreasing form is correct, and the cross-over between them. A target-dependent weight scheme (e.g. a small map from target-signature features to weights, learned end-to-end) is the natural construction if the cross-over turns out to be smooth.

Remark 13 (Single-segment obstruction). A direct computation on the simplest case — one free interior breakpoint (τ, y) on $[0, T]$ with pinned endpoints — shows that exact matching of $\sigma_{1,2}$ alone fixes $y = (2/T) \int_0^T f$ independently of τ , while L^2 -optimality fixes y to a hat-function inner product depending on τ . For $f(t) = \sin(\pi t)$, $T = 1$, symmetry forces $\tau^* = 1/2$ and the values are $y_\sigma = 4/\pi \approx 1.273$ versus $y_{L^2} = 12/\pi^2 \approx 1.216$, a 4.7% discrepancy. The conjecture is therefore not pointwise identity of the two minimisers but norm equivalence in the overdetermined regime.

4.5.4 Concatenation and the frequency-to-level correspondence

By Chen (6) and the linearity of moments, segment-wise regression errors compose into a global error bounded by their sum: for partitions $\{0 = \tau_0 < \dots < \tau_N = T\}$ and per-segment approximations $g^{(i)}$,

$$\sum_{w \in \mathcal{L}_m} w_w (\sigma_w[g] - \sigma_w[X])^2 \leq C(m, N) \sum_{i=1}^N \sum_{w \in \mathcal{L}_m} w_w (\sigma_w[g^{(i)}] - \sigma_w[X^{(i)}])^2, \quad (20)$$

with $C(m, N)$ polynomial in N and exponential in m from the binomial expansions in (7). This is the structural reason why the segment-by-segment inversion of Section 4.3 is consistent with a global signature regression: per-segment errors add additively, and

a target f of characteristic frequency ω_0 needs $N \sim \omega_0 T$ segments of length $\Delta t \sim 1/\omega_0$ to keep the per-segment level requirement bounded. The global L^1 error then scales as $O(N^{-(m+1)})$ for $f \in C^{m+1}$, matching classical piecewise-polynomial approximation rates.

5 Conclusions and forthcoming work

We introduced core signatures, identified them structurally with the Lyndon coordinates of the universal step- n free nilpotent Lie group $G^n(\mathbb{R}^d)$ (Proposition 4 plus Corollary 6), sized them via Witt’s formula (Corollary 7), and gave a constructive Gröbner-basis algorithm for expressing any signature entry in terms of the core. We then used Chen’s identity to invert core signatures into piecewise linear paths of two, three, or more segments, and reframed the multi-segment case as a weighted nonlinear least-squares regression in signature space whose minimiser empirically coincides with the optimal piecewise-linear L^1 approximant. Conjecture 11 states the norm-equivalence result that would make this coincidence rigorous; the moment-matching intuition of Section 4.5 and the single-segment computation give partial support and identify the precise form a proof would need to take.

The application of the core-signature framework to interest-rate term structures and other financial term-structure objects is the subject of a forthcoming paper, in which we apply the inversion to the historical USD Treasury and Brazilian DI futures curves to extract a low-dimensional geometric encoding of the curve dynamics. Two analytical questions left open here are (i) the explicit identification of the Sobolev exponent s and the constants of equivalence in (19), and (ii) the vector-valued generalisation of the moment-matching dictionary required for the term-structure setting where path components are correlated.

The core-signature framework has also been picked up in practitioner work on quantitative-trading futuretesting; see [Bloch (2024)].

We thank Sam Cohen (Oxford) for helpful discussions.

Use of AI assistance. The author used Anthropic’s Claude to review the mathematical content of the 2023 version of this paper, tighten the exposition, supply the theoretical citation chain through the Lyndon basis theorem, Poincaré–Birkhoff–Witt and Radford’s theorem that identifies the core variables with the Lyndon coordinates of $G^n(\mathbb{R}^d)$, and help draft the regression and norm-equivalence framing of Section 4.5. All mathematical claims, the choice of which results to state as conjectures, and the final form of the paper are the author’s responsibility.

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Appendix

A Substitution rules Q_m for $d = 2$

The substitution rules below express every non-core signature entry of $V_1 \cup \dots \cup V_m$ as a polynomial in C_m . The choice of representative within each shuffle orbit is the lex-minimal one (equivalently, the Lyndon representative of each primitive necklace). Q_6 for $d \in \{2, 3\}$ is exported in the companion notebook `CoreSignatures202412.nb`.

The cores referenced below are

$$C_1 = \{s_1, s_2\}, \quad C_2 = C_1 \cup \{s_{1,2}\}, \quad C_3 = C_2 \cup \{s_{1,1,2}, s_{1,2,2}\},$$

$$C_4 = C_3 \cup \{s_{1,1,1,2}, s_{1,1,2,2}, s_{1,2,2,2}\},$$

$$C_5 = C_4 \cup \{s_{1,1,1,1,2}, s_{1,1,1,2,2}, s_{1,1,2,1,2}, s_{1,1,2,2,2}, s_{1,2,1,2,2}, s_{1,2,2,2,2}\}.$$

A.1 Level 2

$s_{1,1}$	$\frac{1}{2}s_1^2$
$s_{2,1}$	$s_1s_2 - s_{1,2}$
$s_{2,2}$	$\frac{1}{2}s_2^2$

A.2 Level 3

$s_{1,1,1}$	$\frac{1}{6}s_1^3$
$s_{1,2,1}$	$s_1s_{1,2} - 2s_{1,1,2}$
$s_{2,1,1}$	$-s_1s_{1,2} + s_{1,1,2} + \frac{1}{2}s_2s_1^2$
$s_{2,1,2}$	$s_2s_{1,2} - 2s_{1,2,2}$
$s_{2,2,1}$	$-s_2s_{1,2} + s_{1,2,2} + \frac{1}{2}s_1s_2^2$
$s_{2,2,2}$	$\frac{1}{6}s_2^3$

A.3 Levels 4 and 5

The level-4 and level-5 substitution rules are listed in the companion notebook `CoreSignatures202412.nb`, sections Q4 and Q5. They have the same structural form as level 3: each non-core entry is a polynomial in core entries of equal or lower level.

B Closed-form solutions for the time-ordered Chen system

B.1 Two-piece system

For the two-piece system of Table 4, dropping equation #5 (Solution 1) gives

$$\{T, t_1, \beta_1, \beta_2\}_{\#1} = \left\{ z_1, \frac{2(z_1z_{1,2} - 3z_{1,1,2})}{z_1(z_1z_2 - 2z_{1,2})}, \beta_1^{(\#1)}, \beta_2^{(\#1)} \right\}, \quad (21)$$

where the closed forms for $\beta_1^{(\#1)}$, $\beta_2^{(\#1)}$ and the Solution-2 expressions are tabulated in the companion notebook `CoreSignatures202412.nb`, section `TwoPiece`. The middle points

$$P_1^{(\#1)} = \left(\frac{2(z_1 z_{1,2} - 3z_{1,1,2})}{z_1 z_2 - 2z_{1,2}}, \frac{-2z_{1,2}(z_1 z_2 - 2z_{1,2}) - 6z_2 z_{1,1,2} + z_1^2 z_2^2}{z_1(z_1 z_2 - 2z_{1,2})} \right), \quad (22)$$

$$P_1^{(\#2)} = \left(\frac{2(3z_1 z_{1,2,2} - 2z_{1,2}^2)}{z_2(z_1 z_2 - 2z_{1,2})}, \frac{z_2(z_1 z_2 - 2z_{1,2}) - 2(z_2 z_{1,2} - 3z_{1,2,2})}{z_1 z_2 - 2z_{1,2}} \right), \quad (23)$$

satisfy the consistency condition (14).

B.2 Three-piece system

The system extends the two-piece equations of Table 4 with three more unknowns ($t_2, \beta_3, \beta_2 \rightarrow \beta'_2, t_1 \rightarrow t'_1$) and four more equations from levels 4 and the cross-terms; the full listing is in the companion notebook `CoreSignatures202412.nb`, section `ThreePiece`. The structure is identical: an over-determined polynomial system whose Gröbner basis admits closed-form solutions for each choice of dropped equation, and whose consistency conditions encode the algebraic limits of the three-segment approximation.

B.3 Beyond three pieces

For four or more segments the algebraic Gröbner approach becomes prohibitively expensive; the regression formulation (15) is the practical alternative. The companion notebook contains the `minim3lines` and `minim4lines` objective functions and the `DifferentialEvolution` configurations used to produce Figures 3, 4, 5–7.